

INTRO TO R SOFTWARE

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Outline

- Why use R
- Flow of data analysis
- R, RStudio, and Posit
- Installation
- Data types in R
- Data structures in R
- Indexing and data wrangling
- Missing values and duplicates
- Descriptive statistics
- Basic plots
- Hands-on
- Some tips

Why use R

- Flexibility to use R everywhere - R is free
- Wide functionalities:
 - Statistical analysis
 - Machine learning, deep learning
 - Image processing (to certain extent)
 - Website development
 - Etc
- Wide range of packages (for example, in [CRAN](https://cran.r-project.org/) - 19,938 packages):
 - Wide range of statistical analysis: basic to advanced
 - Early access to new statistical method
 - All packages are free



R, RStudio, and Posit

- R:
 - Programming and statistical software
 - Run in the back of IDE
- RStudio:
 - Integrated development environment (IDE)
 - Run on front, which the user interact with
- Posit:
 - Previously, also known as RStudio
 - The company that build RStudio

- Other IDEs for R:
 - Google Collab
 - PyCharm
 - VS code - Visual Studio Code
 - Etc
- Why use RStudio:
 - Supported by Posit
 - Probably one of the most (if not the most) popular IDE for R
 - Large support community

Installation

- Posit Cloud:
 - Cloud version of RStudio
 - No need installation
 - [Free up to an extent](#)
 - Need to make an [account](#)
- RStudio:
 - [Install base R first](#)
 - [Install RStudio](#)

Data types in R

- Most common data types in R:
 - Character: “hanis”, “school”
 - Numeric: 1.0, 2.1, 3.0, 3.2
 - Integer: 1, 2, 3, 4L
 - Logical: TRUE, FALSE
 - Complex:
 - Example: $1+4i$
 - Not commonly used
 - Used to store number and an imaginary components

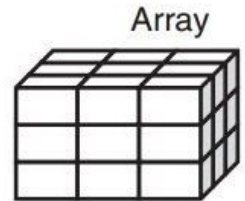
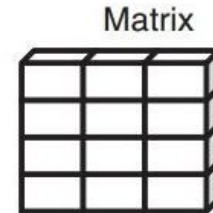
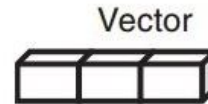
Data structures in R

- Most common data structure in R:
 - Atomic vector:
 - Basic data structure in R
 - Can contain one data type at a time
 - Matrix
 - Array:
 - Matrix but with more than one layer
 - More common in image analysis

Scalar
24

Vector
 $\begin{bmatrix} 2 & -8 & 7 \end{bmatrix}$
row
or
column $\begin{bmatrix} 2 \\ -8 \\ 7 \end{bmatrix}$

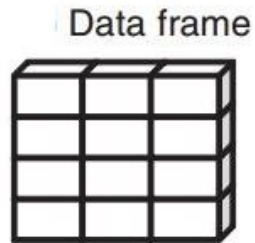
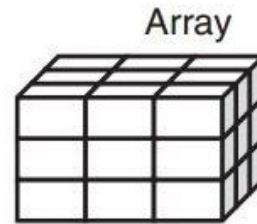
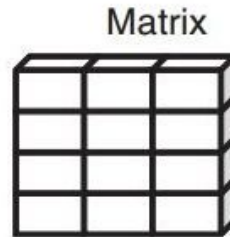
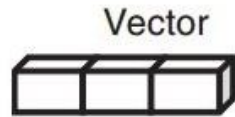
Matrix
 $\begin{bmatrix} 6 & 4 & 24 \\ 1 & -9 & 8 \end{bmatrix}$
row(s) × column(s)



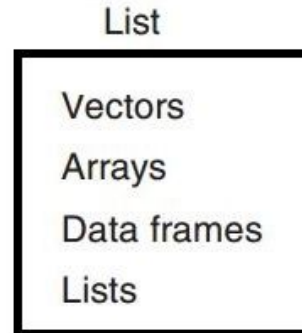
- Data frame
 - Contains the names of variable in the column names and each observation in row
 - Each column may contain different data type
- List
 - May contains different data structures

The diagram illustrates a data frame structure. At the top, the word "Columns" is written in blue. Three blue arrows point from "Columns" to the column headers: "Name", "Team", and "Number". Below the table, the word "Rows" is written in orange. Four orange arrows point from "Rows" to the row indices: 0, 1, 2, and 3. A purple box labeled "Data" is positioned at the bottom right, with purple lines connecting it to the cells containing "8.0", "NaN", "PG", and "NaN" in the "Number", "Team", "Position", and "Age" columns respectively.

	<i>Name</i>	<i>Team</i>	<i>Number</i>	<i>Position</i>	<i>Age</i>
0	Avery Bradley	Boston Celtics	0.0	PG	25.0
1	John Holland	Boston Celtics	30.0	SG	27.0
2	Jonas Jerebko	Boston Celtics	8.0	PF	29.0
3	Jordan Mickey	Boston Celtics	NaN	PF	21.0
4	Terry Rozier	Boston Celtics	12.0	PG	22.0
5	Jared Sullinger	Boston Celtics	7.0	C	NaN
6	Evan Turner	Boston Celtics	11.0	SG	27.0



Columns can be different modes



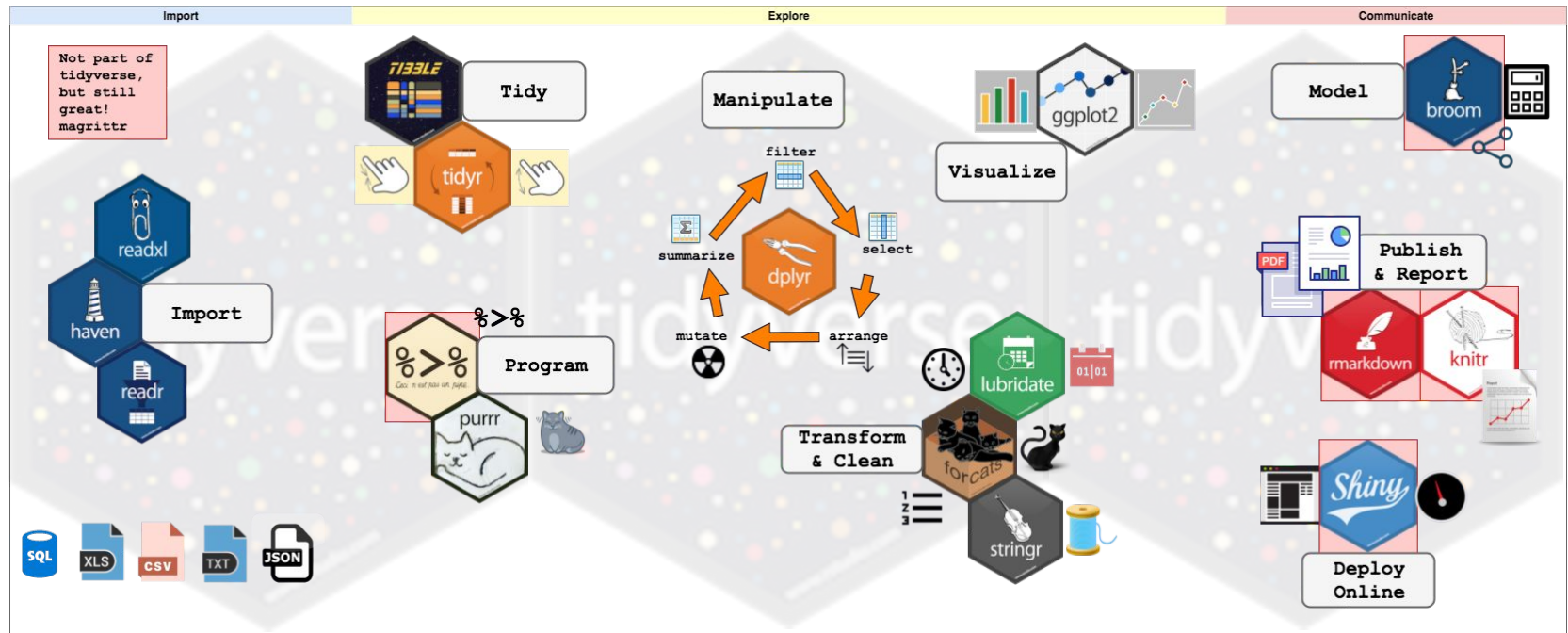
Indexing and data wrangling

- Indexing:
 - Selecting certain elements using their indices
 - This is most commonly done using base R
- Data wrangling:
 - Getting the data into a useful form for visualization, summary, and modeling
 - Can be done using base R and tidyverse approach
 - Tidyverse:
 - A collection of R packages design for data science - developed by Posit team
 - Total of 30 packages with 8 core packages
 - Quite popular among R user especially beginners

Core packages in tidyverse

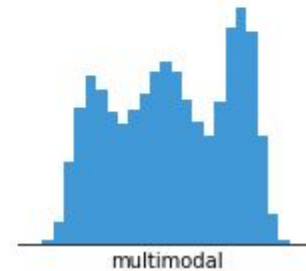
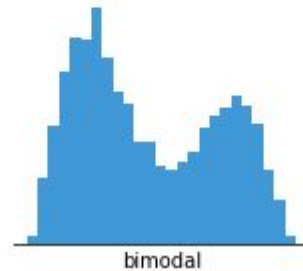
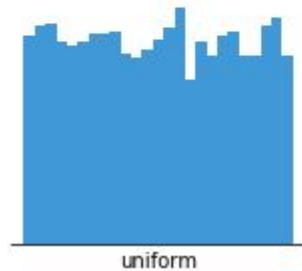
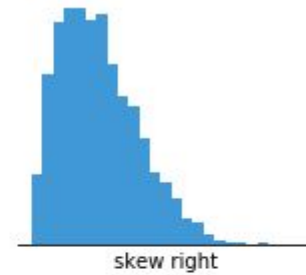
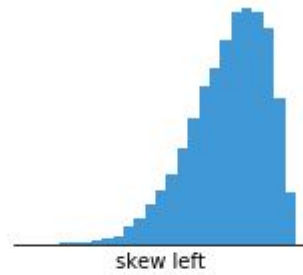
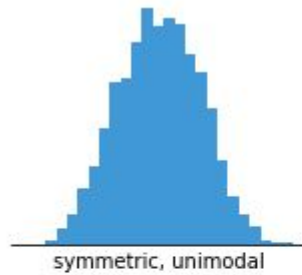


Some common packages in tidyverse



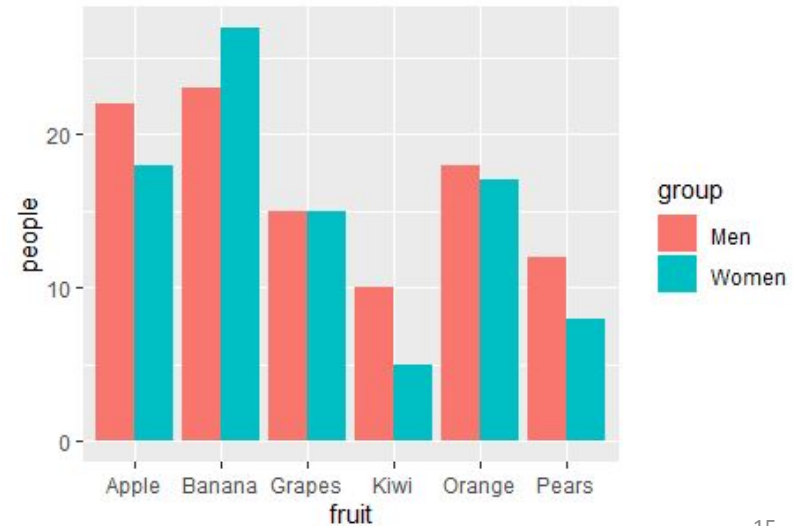
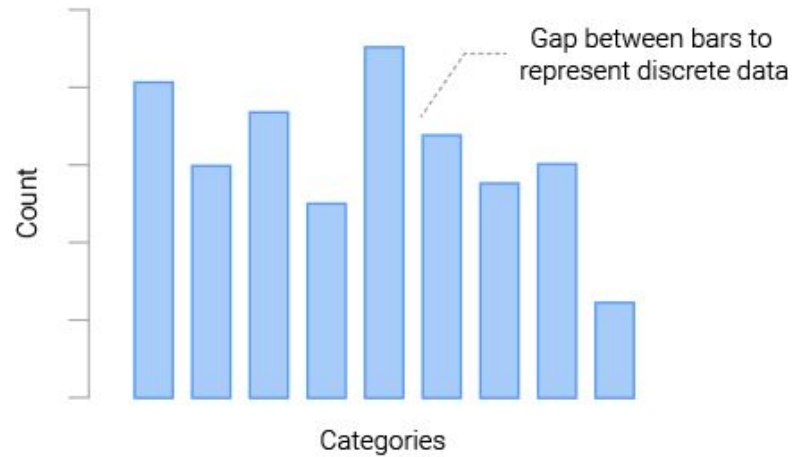
Histogram

- Describe the distribution of the numeric values



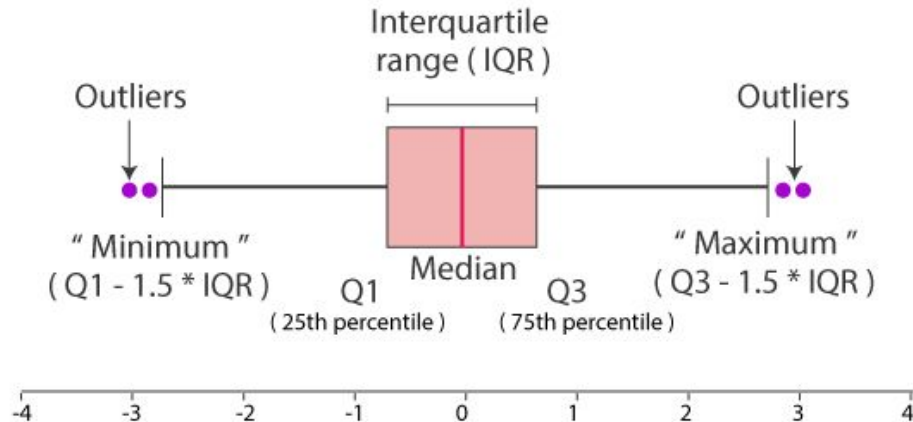
Bar plot

- Describe the distribution of the categorical variable



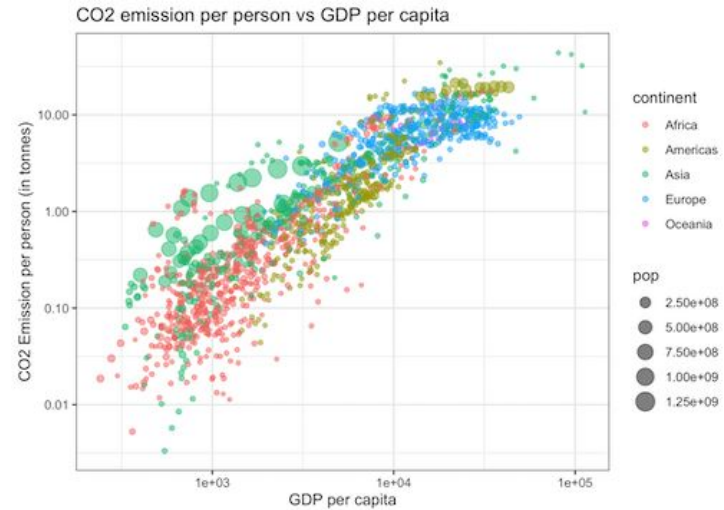
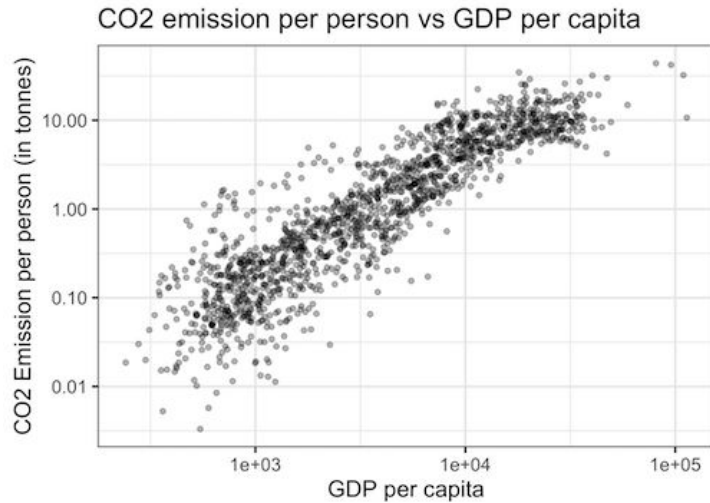
Box plot

- Also known as box and whisker plot
- Can be used to detect outliers - values that deviate too much from other values in the datasets



Scatter plot

- Represent values for two numeric variables



About the datasets - diabetes

- External dataset
- From the National Institute of Diabetes and Digestive and Kidney Diseases
- All patients are females at least 21 years old of Pima Indian heritage
- Link - <https://www.kaggle.com/datasets/mathchi/diabetes-data-set>



- Variables:
 - Pregnancies: Number of times pregnant
 - Glucose: Plasma glucose concentration a 2 hours in an oral glucose tolerance test
 - BloodPressure: Diastolic blood pressure (mm Hg)
 - SkinThickness: Triceps skin fold thickness (mm)
 - Insulin: 2-Hour serum insulin (μ U/ml)
 - BMI: Body mass index ($\text{weight in kg}/(\text{height in m})^2$)
 - DiabetesPedigreeFunction: Diabetes pedigree function
 - Age: Age (years)
 - Outcome: Class variable (0 or 1)

About the datasets - iris

- Dataset available internally in R
- Contains 150 sample of iris flower
- Variables:
 - Species: setosa, versicolor, virginica
 - Sepal.Length
 - Sepal.Width
 - Petal.Length
 - Petal.Width



Virginica



Setosa



Versicolor

About the datasets - fish encounters

- Dataset available internally in R
- Contains information about fish swimming down a river
- Variable:
 - Fish identifier
 - Station - a monitor that record if a fish was seen at that location
 - Seen - was the fish seen (Yes or No)

Hands-on

- Register on Posit Cloud:
 - i. Copy the https link from my GitHub page
 - ii. Log in to Posit Cloud
 - iii. Click “New Project” -> “New Project from Git Repository”
 - iv. Paste the https link in “URL of your Git Repository”
- My GitHub: https://github.com/tengku-hanis/mahtas_workshop.git

Posit Cloud - copy https link

The screenshot shows the GitHub interface for the repository 'jomresearch_intro2r' by user 'tengku-hanis'. The repository is public and has 1 branch and 0 tags. The file list on the left includes 'data', '.gitignore', '1_packages.R', '2_data_types_structures.R', '3_index_wrangling.R', '4_missingData_duplicates.R', '5_descriptive.R', '6_basic_plot.R', '7_exercise.R', 'README.md', and 'intro2r.Rproj'. The 'Code' dropdown menu is open, showing options for cloning the repository via HTTPS, SSH, or GitHub CLI. The HTTPS link is highlighted with a red box and a '2' annotation. The 'Copy' icon next to the link is also highlighted with a red box and a '1' annotation. The README file is selected, showing the title 'jomresearch_intro2r' and the content: 'Material for jomresearch's webinar: Intro to R (for non-coders)' and 'The answers and hints for the exercise is available at https://tengku-hanis.github.io/intro2r_answers/'.

tengku-hanis / jomresearch_intro2r

<> Code Issues Pull requests Actions Projects Wiki Security Insights Settings

jomresearch_intro2r Public

main 1 branch 0 tags

tengku-hanis Commit all material

File	Commit
data	Update R scripts
.gitignore	Initial commit
1_packages.R	Update R scripts
2_data_types_structures.R	Update R script
3_index_wrangling.R	Update R script
4_missingData_duplicates.R	Commit all material
5_descriptive.R	Update R script
6_basic_plot.R	Update R scripts
7_exercise.R	Commit all material
README.md	Commit all material
intro2r.Rproj	Initial commit

Local Codespaces

Clone

HTTPS SSH GitHub CLI

https://github.com/tengku-hanis/jomresearch_intro2r

Use Git or checkout with SVN using the web URL.

Open with GitHub Desktop

Download ZIP

Code 55% faster with AI pair programming.

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1 About

Material for jomresearch's webinar: Intro to R

Readme

Activity

0 stars

1 watching

0 forks

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)

Languages

R 100.0%

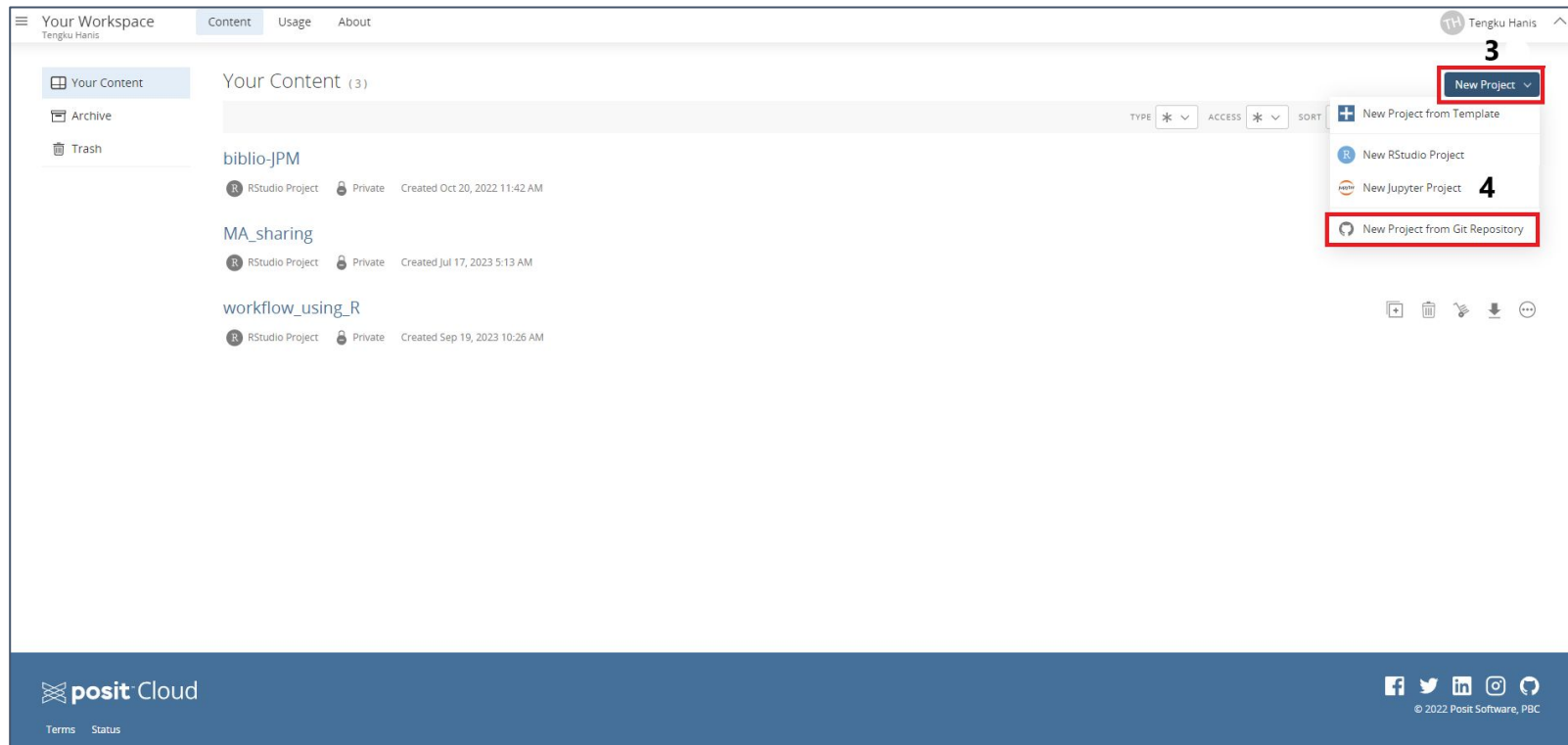
README.md

jomresearch_intro2r

Material for jomresearch's webinar: Intro to R (for non-coders)

The answers and hints for the exercise is available at https://tengku-hanis.github.io/intro2r_answers/

Posit Cloud - log in and click “New Project”



Posit Cloud - paste the https link

The screenshot displays the Posit Cloud web interface. At the top, there's a navigation bar with 'Your Workspace' and 'Tengku Hanis'. Below it, a sidebar on the left shows 'Your Content', 'Archive', and 'Trash'. The main area is titled 'Your Content (3)' and lists three RStudio Projects: 'biblio-JPM' (created Oct 20, 2022), 'MA_sharing' (created Jul 17, 2023), and 'workflow_using_R' (created Sep 19, 2023). A modal dialog box titled 'New Project from Git Repository' is open in the center. It features a 'Project type' dropdown menu currently set to 'RStudio Project', which is highlighted with a red box and the number 5. Below this is a text input field for the 'URL of your Git Repository', also highlighted with a red box and the number 6. An 'OK' button is located at the bottom right of the dialog box. The bottom of the screen shows the Posit Cloud logo and footer information, including social media icons and copyright text.

Tips

- It takes some time before you going to be comfortable and familiar with R
- The more you use R, the faster you become familiar with R





Any question?



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